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1  # SJF Scheduling Program (Non-Preemptive)
2
3  n = int(input("Enter number of processes: "))
4
5  processes = []
6
7  # Input process details
8  for i in range(n):
9      at = int(input(f"Enter arrival time for process {i+1}: "))
10     bt = int(input(f"Enter burst time for process {i+1}: "))
11     processes.append({
12         "pid": i + 1,
13         "arrival": at,
14         "burst": bt,
15         "completed": False,
16         "waiting": 0,
17         "turnaround": 0
18     })
19
20     time = 0
21     completed = 0
22
23     while completed < n:
24         # Get all processes that have arrived and not completed
25         ready_queue = [p for p in processes if p["arrival"] <= time and not p["completed"]]
26
27         if ready_queue:
28             # Select process with smallest burst time
29             current = min(ready_queue, key=lambda x: x["burst"])

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27         if ready_queue:
28             # Select process with smallest burst time
29             current = min(ready_queue, key=lambda x: x["burst"])
30
31             # Calculate times
32             current["waiting"] = time - current["arrival"]
33             time += current["burst"]
34             current["turnaround"] = current["waiting"] + current["burst"]
35
36             current["completed"] = True
37             completed += 1
38         else:
39             # If no process is ready, move time forward
40             time += 1
41
42     # Display results
43     print("\nPID\tAT\tBT\tWT\tTAT")
44     for p in processes:
45         print(f"{p['pid']}\t{p['arrival']}\t{p['burst']}\t{p['waiting']}\t{p['turnaround']}")
46
47     # Calculate averages
48     avg_wt = sum(p["waiting"] for p in processes) / n
49     avg_tat = sum(p["turnaround"] for p in processes) / n
50
51     print(f"\nAverage Waiting Time: {avg_wt:.2f}")
52     print(f"Average Turnaround Time: {avg_tat:.2f}")

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Enter number of processes: 4
Enter arrival time for process 1: 8
Enter burst time for process 1: 9
Enter arrival time for process 2: 8
Enter burst time for process 2: 7
Enter arrival time for process 3: 9
Enter burst time for process 3: 6
Enter arrival time for process 4: 7
Enter burst time for process 4: 8
```

PID	AT	BT	WT	TAT
1	8	9	20	29
2	8	7	13	20
3	9	6	6	12
4	7	8	0	8

Average Waiting Time: 9.75

Average Turnaround Time: 17.25